Utthan Singh Roy

Roll No.:23035010081
Online BSc (Hons.) - Data Science and AI
Indian Institute of Technology, Guwahati

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Education

Indian Institute of Technology, Guwahati — *2023 – Present*
B.Sc. (Hons.) in Data Science and Artificial Intelligence | CGPA: 8.34 (Current) — Guwahati, Assam

Madan Mohan Malaviya University of Technology — *2023 – Present*
B.Tech in Electrical Engineering | CGPA: 8.36 (Current) — Gorakhpur, UP

St. Joseph's School (CBSE) — *2019 – 2021*
Senior Secondary (Class XII): 95.4% | Secondary (Class X): 93.8% — Shaktinagar, UP

Experience

Indian Institute of Technology (IIT) Guwahati — *Jul. 2026 – Present*
LLM Application Engineering Intern — Guwahati, Assam

- Architected a low-latency Hybrid RAG pipeline combining ChromaDB/BM25 retrieval with asynchronous LCEL token streaming to minimize Time-to-First-Token (TTFT).
- Engineered citation and deterministic fallback guardrails for LLM outputs, improving reliability of production-oriented microservice responses.

Projects

AI-Powered Market Regime Shift Classifier — *Feb. 2026 – Jun. 2026*
Python, XGBoost, FinBERT, Streamlit, GitHub Actions — GitHub

- Engineered a cost-sensitive XGBoost model on 41 quantitative and FinBERT sentiment features, achieving 0.97 ROC-AUC for systemic market stress detection.
- Built a real-time inference dashboard and automated feature engineering, deployment, and model updates through GitHub Actions CI/CD.

Physics-Validated AI Architecture for Autonomous Grid Resilience — *Apr. 2026 – Jul. 2026*
Python, LightGBM, SAC Reinforcement Learning, LangGraph — GitHub

- Designed Soft Actor-Critic (SAC) reinforcement learning policies with a LightGBM forecasting layer achieving 1.43% MAPE and 100% convergence in AC power-flow simulations.
- Implemented a deterministic validation firewall for LLM-generated targets, dynamically rejecting invalid outputs and improving the Grid Resilience Index by 6.41 points.

Automated Sustainable Portfolio Optimizer — *Oct. 2025 – Jan. 2026*
Python, XGBoost, NSGA-II, GitHub Actions, Pandas — GitHub

- Combined NSGA-II optimization with XGBoost to forecast 30-day Nifty 50 directionality and dynamically optimize asset weights under tail-risk constraints (CVaR).
- Automated daily data ingestion, model retraining, and deployment through a GitHub Actions pipeline.

Technical Skills

Programming: Python, SQL, C, Java*, R*

AI/ML: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, SAC, NSGA-II, FinBERT, SHAP*

LLM/Agentic AI: LangGraph, RAG, LCEL, ChromaDB, BM25, API Integration

MLOps/Tools: Git, GitHub Actions, Streamlit, SQLite, Pandas, NumPy

** Elementary proficiency*

Key Courses Taken

Computer Science & AI: Data Structures, Algorithm Design & Analysis, Deep Learning for Computer Vision, Hardware-Aware Deep Learning, LLM Application Engineering

Mathematics & Analytics: Linear Algebra, Optimization, Statistical Inferencing, Time Series Analysis

Systems & Engineering: Control System Engineering, Digital Control System, Microprocessors

Positions of Responsibility

Volunteer, AI Confluence, IIT Guwahati — *2024*

Certifications & Achievements

Explainable AI (XAI) Specialization, Coursera, Duke University — *2026*

AI in Healthcare Specialization, Coursera, Stanford University — *2026*

Qualified GATE 2026, Data Science & Artificial Intelligence (DA) — *2026*